

Hack The Change- Virtual Quake Dampening for Earthquake Mitigation

1. TASK DEFINITION

This project aims to develop a simulation framework to evaluate and improve the resilience of multi-story buildings using real earthquake data and AI-driven optimization. The tool queries the USGS API to retrieve earthquake metadata, including magnitude, depth, and geolocation which then generates a realistic synthetic ground motion time history using a simplified ground motion prediction model. This simulated ground acceleration is applied to a nonlinear multi-degree-of-freedom (MDOF) lumped-mass structural model representing the building.

Each floor of the building is modeled as a lumped mass connected by nonlinear story springs and dampers that approximate stiffness, yielding, and energy dissipation. The system dynamically solves the nonlinear equilibrium equation, as defined by equation 1 in the Appendix of this paper.

1.1 Target

The target of this project is to create an AI-assisted simulation tool that can be implemented before the construction process to predict how buildings would react during earthquakes. By using ground motion data from the USGS API, a series of transformative equations were applied to find how shaking moves each floor of the building. The AI then uses these metrics to recommend improvements that would mitigate the risk of destruction during earthquakes.

1.2 Significance

The aim of this project is to help engineers and city planners to create better infrastructure through better building designs that will withstand the impacts of catastrophic earthquakes.

This project drew inspiration from the series of devastating earthquakes that struck Mexico in 2017. Within a twelve-day span two separate earthquakes struck the country. The first being an earthquake in Mexico City with a magnitude of 7.1, then another in a town only 75 miles from Mexico City called Puebla, with a magnitude of 8.2. Overall, 184,000 homes and 16,000 schools were destroyed. The earthquakes claimed the lives of 370 people. The World Bank provided a \$150 million fund for Mexico to use in order to accommodate to the survivors left homeless [1].

This was not the first devastating earthquake to hit Mexico. In 1985, an 8.0 magnitude earthquake struck Mexico City, destroying 600 homes, killing 10,000 people, and leaving 250,000 homeless. Once again, the World Bank provided the funds to help rebuild almost 100,000 new homes. Many have questioned whether Mexico City learned anything from past disasters, as many of the destroyed buildings had been built only 9 to 12 months prior, exposing a lack of disaster preparedness [2].

This issue is not unique to Mexico City, as many other mega cities such as Tokyo, Istanbul, and Manila face the same issues. These cities contain a high population density, and high magnitude earthquakes often leave thousands displaced. This creates a global issue related to home insecurity and risks the lives of millions.

Similar existing tools include Extreme Loading for Structures (ELS) software, which is designed for structural analysis. However, our solution implements AI-driven optimization that can be implemented prior to the construction of a building to mitigate risk [3].

1.3 Background

ELS software uses Applied Element Method (AEM), which models structures as networks of interconnected elements for separate analysis under load. This approach provides highly detailed insight into local damage mechanisms, but it requires intensive computational resources. Because the lumped mass method treats each floor as a single mass connected by springs and dampers, it reduces the degrees of freedom, and instead of solving many nonlinear interactions between individual elements, the model solves a small set of non-linear dynamic equations using time-integration methods. As a result, users can input parameters, run simulations, and visualize earthquake responses in real time through a web interface, which makes the tool interactive and accessible. This contrasts with nonlinear methods like AEM, which can take hours to process a single detailed simulation due to iterative cracking and contact calculations.

The Richter scale was the first logarithmic measure of earthquake size and is based on high-frequency seismic data from local stations. As global seismic networks expanded, it is clear that the Richter method was only valid for nearby, moderate-sized earthquakes. New scales such as body-wave magnitude (M_b) and surface-wave magnitude (M_s) are introduced, each tailored to specific frequency ranges. However, these scales saturate for very large earthquakes. To overcome this the moment magnitude scale (M_w) was developed, which is derived from the seismic moment (the product of the fault area, average slip, and rock rigidity), representing the total energy released by an earthquake. Moment magnitude provides a consistent, physically based measure of earthquake size across all magnitudes, making it the standard used by the USGS and in modern seismology [4].

2.METHODOLOGY

2.1 Data Input Strategy

The USGS monitors earthquakes globally by recording seismic waveforms from multiple locations. From that data, they compile metadata, which includes origin time, latitude, longitude, depth, amplitude, wave arrival times, and frequency content. The USGS data is set to calculate moment magnitude from the seismic moment, which can be found using the aforementioned parameters.

Our script utilizes this given M_w value from the USGS metadata to compute the event size by putting filters on the API, such as minimum & maximum latitude and longitude, minimum magnitude, and a time window. This ensures that the dataset focuses on seismic events relevant to the region of interest and is of significant size, to simulate building response. We set a minimum magnitude threshold of 5.0 in the USGS query to exclude minor earthquakes that do not produce measurable structural effects. This decision was made based on the Modified Mercalli Intensity Scale [5]. This scale along with the equations used to find moment magnitude can be found in the Appendix of this report.

Each selected event based on our threshold provides the basis for the ground acceleration time series, which is applied to our lumped-mass structural model. This model works by connecting the different aspects of the building together, such as mass, stiffness and damping elements.

2.2 Setup

Once the event has been selected, we estimated the ground motion time history for a specific geolocation. This is done by creating an acceleration array that is found using the ground motion prediction to estimate PGA, duration and dominant frequency, and motion type. This array is what is used to generate the structural model. Using the Newmark- β integration method to compute displacement, velocity, and acceleration responses for each floor over time, these results are post-processed to calculate

structural performance metrics such as maximum inter-story drift, peak floor acceleration, base shear, and overall damage index. This is done using the Multiple Degrees of Freedom (MDOF) systems equation. Using inputs such as mass, damping, and stiffness matrices are calculated against time to find the displacement, velocity and acceleration values needed to calculate the damage index.

Once we have computed the drift, peak floor acceleration, and damage index metrics, they are combined with the design parameters and sent to the OpenAI API prompt to generate optimized design suggestions.

2.2 Relevant Terminology

Braced Frame: Vertical truss in a building frame system to resist seismic forces [8].

Moment Frame: A frame in which members resist lateral reaction forces by rotation along the axis of the members [8].

Base Shear: Total design lateral force or shear at the base [8].

S_d (Spectral Displacement): Maximum relative displacement of the oscillator [8].

S_v (Spectral Velocity): Maximum relative velocity response [8].

S_a (Spectral Acceleration): Maximum acceleration experienced by the mass [8].

C_s: The ratio of the base shear force to the total building weight, which is often a unitless coefficient with a value between 0.05-0.3 [8].

3. Evaluation Criteria

3.1 Evaluation Metrics

To see if our simulation is realistic, we looked at key numbers that show how the building moves

and how much damage it might take. The main ones are inter-story drift, peak floor acceleration, and a simple damage index that tells how close the structure is to failure. These results help us tell whether the design is stable and if the AI's changes are actually improving the building's performance. Each design suggestion can be traced back to specific simulation outcomes, allowing users to understand why a recommendation was made.

3.2 Fairness Metrics

We made sure our AI model stays fair and consistent when giving design suggestions. Since the AI looks at numbers like drift, acceleration, and damage, we checked that it gives similar advice for buildings with similar results and doesn't just suggest expensive fixes that only rich cities could afford. We also wanted to make sure the suggestions make sense for local materials and building styles, so the tool could work just as well in developing areas as in major cities. This helps keep the system more balanced and usable in different regions.

4. References

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4. Appendix

$$[M]\{\ddot{u}(t)\} + [C]\{\dot{u}(t)\} + [K]\{u(t)\} = -[M]\{1\}\ddot{u}_g(t) \quad (1)$$

where $[M]$, $[C]$, and $[K]$ are the mass, damping, and stiffness matrices; $\{u(t)\}$, $\{\dot{u}(t)\}$, and $\{\ddot{u}(t)\}$ are the displacement, velocity, and acceleration vectors of the floors; and $\ddot{u}_g(t)$ is the ground acceleration input.

$$[K] = k_{\text{base}} + k_{\text{story}} - k_{\text{story}} \quad (2)$$

$$[C] = 2\xi\omega_n[M] \quad (3)$$

where ξ is the damping ratio and $\omega_n = \sqrt{k_{\text{story}}/m}$ is the natural frequency.

$$\begin{aligned} [K_{eff}] &= [K] + \frac{\gamma}{\beta\Delta t}[C] + \frac{1}{\beta\Delta t^2}[M] \\ \{p_{eff}\} &= -[M]\{1\}\ddot{u}_g(t_i) + [M]\left(\frac{1}{\beta\Delta t^2}\{u_{i-1}\} + \frac{1}{\beta\Delta t}\{\dot{u}_{i-1}\} + \left(\frac{1}{2\beta} - 1\right)\{\ddot{u}_{i-1}\}\right) \\ &+ [C]\left(\frac{\gamma}{\beta\Delta t}\{u_{i-1}\} + \left(\frac{\gamma}{\beta} - 1\right)\{\dot{u}_{i-1}\} + \Delta t\left(\frac{\gamma}{2\beta} - 1\right)\{\ddot{u}_{i-1}\}\right) \\ \{u_i\} &= [K_{eff}]^{-1}\{p_{eff}\} \\ \{\dot{u}_i\} &= \frac{1}{\beta\Delta t}(\{u_i\} - \{u_{i-1}\}) - \frac{1}{\beta\Delta t}\{\dot{u}_{i-1}\} - \left(\frac{1}{2\beta} - 1\right)\{\ddot{u}_{i-1}\} \\ \{\ddot{u}_i\} &= \{\dot{u}_{i-1}\} + \Delta t[(1-\gamma)\{\ddot{u}_{i-1}\} + \gamma\{\ddot{u}_i\}] \end{aligned}$$

where $\beta = 0.25$ and $\gamma = 0.5$ correspond to the average-acceleration Newmark method.

$$IDR_i = \frac{|u_i - u_{i-1}|}{h_i} \quad (4)$$

Inter-story Drift Ratio (IDR) measures how much a building bends during the earthquake by calculating how much one floor sways relative to the floor below it, divided by the height between the floors.

$$a_{peak,i} = \max_t |\ddot{u}_i(t)| \quad (5)$$

Peak Floor Acceleration (*apeak*) is the greatest amount of acceleration a floor in the building experiences during the earthquake.

$$D = \min\left(1, \frac{\max(IDR_i)}{0.02}\right) \quad (6)$$

Modified Mercalli Intensity Scale:

1: Felt by very few people; barely noticeable.

5: Felt by many indoors, few outdoors; may feel like a heavy truck passing by.

7: Slight to moderate damage in well-built structures; considerable damage to poorly built ones. Some walls may fall.

10: Most masonry and frame structures destroyed; ground badly cracked, widespread destruction and landslides.

12: Total destruction; ground waves visible, objects thrown into the air.

